

SOLIDWORKS API (.NET VBA C#) — Reference (320/337): IVersionCompatibilityItem-IVertex

SOLIDWORKS API Documentation (converted from CHM)

IVersionCompatibilityItem (Type)

IVersionCompatibilityItem Interface

[See Also](#) [Members](#)



[SolidWorks.Interop.sldworks Namespace](#) : IVersionCompatibilityItem Interface

Describes a reason a document cannot be saved to the file format of a previous version of SOLIDWORKS.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface IVersionCompatibilityItem
```

Visual Basic (Usage)



```
Dim instance As IVersionCompatibilityItem
```

C#

```
public interface IVersionCompatibilityItem
```

C++/CLI

```
public interface class IVersionCompatibilityItem
```

Access Diagram

[VersionCompatibilityItem](#)

See Also

[IVersionCompatibilityItem Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

IVersionCompatibilityItem Interface Members

[See Also](#) [Properties](#)



[SolidWorks.Interop.sldworks Namespace](#) : IVersionCompatibilityItem Interface

The following tables list the members exposed by [IVersionCompatibilityItem](#).

Public Properties

Name	Description
 Action	Gets the localized message describing he corrective action recommended for this incompatibility.
 ID	Gets the message identifier for this incompatibility.
 IncompatibleObject	Get the object incompatible with the specified file format version.
 Message	Gets the localized message describing this incompatibility.
 Version	Gets the name of the version of SOLIDWORKS to which this incompatibility relates.

[Top](#)

See Also

[IVersionCompatibilityItem Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

IVersionCompatibilityItem Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IVersionCompatibilityItem Interface

For a list of all members of this type, see [IVersionCompatibilityItem members](#).

Public Properties

Name	Description
 Action	Gets the localized message describing the corrective action recommended for this incompatibility.
 ID	Gets the message identifier for this incompatibility.
 IncompatibleObject	Get the object incompatible with the specified file format version.
 Message	Gets the localized message describing this incompatibility.
 Version	Gets the name of the version of SOLIDWORKS to which this incompatibility relates.

[Top](#)

See Also

[IVersionCompatibilityItem Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

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Action Property (IVersionCompatibilityItem)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IVersionCompatibilityItem Interface](#) : Action Property (IVersionCompatibilityItem)

Gets the localized message describing the corrective action recommended for this incompatibility.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property Action As System.String

Visual Basic (Usage)



```
Dim instance As IVersionCompatibilityItem  
Dim value As System.String
```

```
value = instance.Action
```

C#

```
System.string Action {get;}
```

C++/CLI

```
property System.String^ Action {  
    System.String^ get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [VersionCompatibilityItem::Action](#).

See Also

[IVersionCompatibilityItem Interface](#)
[IVersionCompatibilityItem Members](#)

Availability

SOLIDWORKS 2026 SP03, Revision Number 34.3

ID Property (IVersionCompatibilityItem)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [IVersionCompatibilityItem Interface](#) : ID Property (IVersionCompatibilityItem)

Gets the message identifier for this incompatibility.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property ID As System.String

Visual Basic (Usage)



```
Dim instance As IVersionCompatibilityItem
Dim value As System.String

value = instance.ID
```

C#

```
System.string ID {get;}
```

C++/CLI

```
property System.String^ ID {
    System.String^ get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [VersionCompatibilityItem.ID](#).

See Also

[IVersionCompatibilityItem Interface](#)
[IVersionCompatibilityItem Members](#)

Availability

SOLIDWORKS 2026 SP03, Revision Number 34.3

IncompatibleObject Property (IVersionCompatibilityItem)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [IVersionCompatibilityItem Interface](#) : IncompatibleObject Property (IVersionCompatibilityItem)

Get the object incompatible with the specified file format version.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property IncompatibleObject As System.Object

Visual Basic (Usage)



```
Dim instance As IVersionCompatibilityItem
Dim value As System.Object

value = instance.IncompatibleObject
```

C#

```
System.Object IncompatibleObject {get;}
```

C++/CLI

```
property System.Object^ IncompatibleObject {  
    System.Object^ get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [VersionCompatibilityItem::IncompatibleObject](#).

Remarks

The incompatible object is typically a [feature](#).

See Also

[IVersionCompatibilityItem Interface](#)
[IVersionCompatibilityItem Members](#)

Availability

SOLIDWORKS 2026 SP03, Revision Number 34.3

SOLIDWORKS API Help

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Message Property (IVersionCompatibilityItem)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IVersionCompatibilityItem Interface](#) : Message Property (IVersionCompatibilityItem)

Gets the localized message describing this incompatibility.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property Message As System.String
```

Visual Basic (Usage)



```
Dim instance As IVersionCompatibilityItem  
Dim value As System.String
```

```
value = instance.Message
```

C#

```
System.string Message {get;}
```

C++/CLI

```
property System.String^ Message {  
    System.String^ get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [VersionCompatibilityItem::Message](#).

See Also

Availability

SOLIDWORKS 2026 SP03, Revision Number 34.3

SOLIDWORKS API Help

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Version Property (IVersionCompatibilityItem)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IVersionCompatibilityItem Interface](#) : Version Property (IVersionCompatibilityItem)

Gets the name of the version of SOLIDWORKS to which this incompatibility relates.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property Version As System.String

Visual Basic (Usage)



```
Dim instance As IVersionCompatibilityItem  
Dim value As System.String
```

```
value = instance.Version
```

C#

```
System.string Version {get;}
```

C++/CLI

```
property System.String^ Version {  
    System.String^ get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [VersionCompatibilityItem::Version](#).

Remarks

The string returned is the year element of the SOLIDWORKS version name. I.e. "2024" or "2025" or "2026", etc.

See Also

[IVersionCompatibilityItem Interface](#)
[IVersionCompatibilityItem Members](#)

Availability

SOLIDWORKS 2026 SP03, Revision Number 34.3

IVertex (Type)

SOLIDWORKS API Help

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IVertex Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IVertex Interface

Represents the start or end of an [edge](#).

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface IVertex
```

Visual Basic (Usage)



```
Dim instance As IVertex
```

C#

```
public interface IVertex
```

C++/CLI

```
public interface class IVertex
```

Visual Basic for Applications (VBA) Syntax

See [Vertex](#).

Example

[Display Vertices \(VBA\)](#)
[Get Maximum Edge and Vertex Gaps \(VBA\)](#)
[Get Maximum Edge and Vertex Gaps \(VB.NET\)](#)
[Get Maximum Edge and Vertex Gaps \(C#\)](#)

Remarks

This interface allows you to get the associated edges in an enumerated list or the coordinates of the vertex.

Accessors

[IAdvancedHoleElementData::SelectionEntity](#)
[IBeltChainFeatureData::BeltLocationPlane](#)
[IBody2::AddVertexPoint](#) and [IBody2::!AddVertexPoint](#)
[IBody2::GetVertices](#) and [IBody2::!GetVertices](#)
[ICamFollowerMateFeatureData::EntitiesToMate](#)
[IChainPatternFeatureData::Group1PathLink1](#)
[IChainPatternFeatureData::Group1PathLink2](#)
[IChainPatternFeatureData::Group2PathLink1](#)
[IChainPatternFeatureData::Group2PathLink2](#)
[IChamferFeatureData2::Vertex](#) and [IChamferFeatureData2::!Vertex](#)
[ICoincidentMateFeatureData::EntitiesToMate](#)
[IConcentricMateFeatureData::EntitiesToMate](#)
[IEdge::GetEndVertex](#) and [IEdge::!GetEndVertex](#)
[IEdge::GetStartVertex](#) and [IEdge::!GetStartVertex](#)
[IFacet::GetFacetVertices](#)

[FillPatternFeatureData::SeedFeatureCenter](#)
[IHingeMateFeatureData::EntitiesToMate](#)
[IIndentFeatureData::ToolBodyRegion](#)
[ILinearPatternFeatureData::D1EndReference](#)
[ILinearPatternFeatureData::D1EndSeedReference](#)
[ILinearPatternFeatureData::D2EndReference](#)
[ILinearPatternFeatureData::D2EndSeedReference](#)
[ILocalCurvePatternFeatureData::SelectedPoint](#)
[ILocalLinearPatternFeatureData::D1EndReference](#)
[ILocalLinearPatternFeatureData::D1EndSeedReference](#)
[ILocalLinearPatternFeatureData::D2EndReference](#)
[ILocalLinearPatternFeatureData::D2EndSeedReference](#)
[ILoop2::GetVertices](#) and [ILoop2::IGetVertices](#)
[IModeler::ImprintingFaces](#) and [IModeler::IImprintingFaces](#)
[IMoveCopyBodyFeatureData::TransofrmReferenceEntity](#)
[IMoveCopyBodyFeatureData::TranslateToVertex](#)
[IMoveFaceFeatureData::GetEndConditionEntity](#)
[IPrimaryMemberPointLengthFeatureData::GetPoints](#)
[ISecondaryMemberUpToMembersFeatureData::GetFromPoint](#)
[ISecondaryMemberUpToMembersFeatureData::GetPointMemberPairs](#)
[ISimpleFilletFeatureData2::GetSetbackVertices](#) and [ISimpleFilletFeatureData2::IGetSetbackVertices](#)
[ISimpleFilletFeatureData2::HelpPoint](#)
[ISimpleHoleFeatureData2::Vertex](#) and [ISimpleHoleFeatureData2::IVertex](#)
[ISketchPatternFeatureData::ReferencePoint](#)
[ISMGussetFeatureData::ReferencePoint](#)
[ISurfaceExtendFeatureData::Vertex](#)
[ISurfaceFlattenFeatureData::FixPointVertex](#)
[ISurfExtrudeFeatureData::GetVertex](#) and [ISurfExtrudeFeatureData::IGetVertex](#)
[ISweepFeatureData::EndDirectionVector](#)
[ISweepFeatureData::StartDirectionVector](#)
[ISymmetricMateFeatureData::EntitiesToMate](#)
[ITabAndSlotGroupData::SelectionEndReferencePoint](#)
[ITabAndSlotGroupData::SelectionStartReferencePoint](#)
[ITablePatternFeatureData::ReferencePoint](#)
[IThreadFeatureData::GetEndConditionReference](#)
[IThreadFeatureData::ProfileLocation](#)
[IThreadFeatureData::StartEntity](#)
[IUniversalJointMateFeatureData::JointPoint](#)
[IVariableFilletFeatureData2::GetSetbackVertices](#) and [IVariableFilletFeatureData2::IGetSetbackVertices](#)
[IWizardHoleFeatureData2::Vertex](#) and [IWizardHoleFeatureData2::IVertex](#)

Access Diagram

[Vertex](#)

See Also

[IVertex Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

[See Also](#) [Methods](#)

[SolidWorks.Interop.sldworks Namespace](#) : IVertex Interface

The following tables list the members exposed by [IVertex](#).

 Public Methods

Name	Description
 Display	Highlights the vertex in the specified color.
 EnumEdges	Gets a list of enumerated edges corresponding to this vertex.
 EnumEdgesOriented	Gets the enumerated edges and orients them per coedge in the direction corresponding to this vertex.
 GetAdjacentFaces	Gets the faces adjacent to this vertex.
 GetClosestPointOn	Gets the closest point on the vertex using the X, Y, Z input point.
 GetEdges	Gets a list of enumerated edges corresponding to this vertex.
 GetEdgesOriented	Gets the enumerated edges and orients them per coedge in the direction corresponding to this vertex.
 GetPoint	Gets the point corresponding to this vertex.
 GetTrackingIDs	Gets the tracking IDs assigned to this vertex .
 GetTrackingIDsCount	Gets the number of tracking IDs assigned to this vertex.
 IGetAdjacentFaces	Gets the faces adjacent to this vertex.
 IGetAdjacentFacesCount	Gets the number of faces adjacent to this vertex.
 IGetClosestPointOn	Gets the closest point on the vertex using the X, Y, Z input point.
 IGetPoint	Gets the point corresponding to this vertex.
 IGetTrackingIDs	Gets the tracking IDs assigned to this vertex .
 IsTolerant	Gets whether a vertex is tolerant and its tolerance value.
 RemoveTrackingID	Removes a tracking ID assigned to this vertex .
 SetTrackingID	Assigns a tracking ID to this vertex.

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 See Also

[IVertex Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IVertex Interface Methods

[See Also](#)

[SolidWorks.Interop.sldworks Namespace](#) : IVertex Interface

For a list of all members of this type, see [IVertex members](#).

 Public Methods

Name	Description
 Display	Highlights the vertex in the specified color.
 EnumEdges	Gets a list of enumerated edges corresponding to this vertex.
 EnumEdgesOriented	Gets the enumerated edges and orients them per coedge in the direction corresponding to this vertex.
 GetAdjacentFaces	Gets the faces adjacent to this vertex.
 GetClosestPointOn	Gets the closest point on the vertex using the X, Y, Z input point.
 GetEdges	Gets a list of enumerated edges corresponding to this vertex.
 GetEdgesOriented	Gets the enumerated edges and orients them per coedge in the direction corresponding to this vertex.
 GetPoint	Gets the point corresponding to this vertex.
 GetTrackingIDs	Gets the tracking IDs assigned to this vertex .

		
	GetTrackingIDsCount	Gets the number of tracking IDs assigned to this vertex.
	IGetAdjacentFaces	Gets the faces adjacent to this vertex.
	IGetAdjacentFacesCount	Gets the number of faces adjacent to this vertex.
	IGetClosestPointOn	Gets the closest point on the vertex using the X, Y, Z input point.
	IGetPoint	Gets the point corresponding to this vertex.
	IGetTrackingIDs	Gets the tracking IDs assigned to this vertex .
	IsTolerant	Gets whether a vertex is tolerant and its tolerance value.
	RemoveTrackingID	Removes a tracking ID assigned to this vertex .
	SetTrackingID	Assigns a tracking ID to this vertex.

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See Also

[IVertex Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

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Display Method (IVertex)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IVertex Interface](#) : Display Method (IVertex)

Highlights the vertex in the specified color.

.NET Syntax

Visual Basic (Declaration)

```
Sub Display( _
    ByVal TopDoc As ModelDoc2, _
    ByVal Color As System.Integer, _
    ByVal Scale As System.Double, _
    ByVal HighlightState As System.Boolean _
)
```

Visual Basic (Usage)



```
Dim instance As IVertex
Dim TopDoc As ModelDoc2
Dim Color As System.Integer
Dim Scale As System.Double
Dim HighlightState As System.Boolean

instance.Display(TopDoc, Color, Scale, HighlightState)
```

C#

```
void Display(
    ModelDoc2 TopDoc,
    System.int Color,
    System.double Scale,
    System.bool HighlightState
)
```

C++/CLI

```
void Display(
    & ModelDoc2^ TopDoc,
    & System.int Color,
    & System.double Scale,
    & System.bool HighlightState
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TopDoc
[Model](#) in which to display the vertex
Color
COLORREF value for highlighting
Scale
Radius of the circle used to display the vertex

NOTE: Vertex is displayed as a circle. By default, the radius is 4 pixels. Therefore, a scale of 1 is equal to 4 pixels.

HighlightState
True to highlight the vertex, false to not

 Visual Basic for Applications (VBA) Syntax

See [Vertex::Display](#).

 Example

[Display Vertices \(VBA\)](#)

 Remarks

If HighlightState set to...	Then the vertex is...
True	Highlighted in the color specified for Color
False	Hidden and Color is ignored

 See Also

[IVertex Interface](#)
[IVertex Members](#)

 Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help [Send comments on this topic.](#)

EnumEdges Method (IVertex)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IVertex Interface](#) : EnumEdges Method (IVertex)

Gets a list of enumerated edges corresponding to this vertex.

 .NET Syntax

Visual Basic (Declaration)

Function EnumEdges() As EnumEdges

Visual Basic (Usage)



```
Dim instance As IVertex  
Dim value As EnumEdges  
  
value = instance.EnumEdges()
```

C#

EnumEdges EnumEdges()

C++/CLI

```
EnumEdges^ EnumEdges();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[Enumerated edges](#)

Visual Basic for Applications (VBA) Syntax

See [Vertex::EnumEdges](#).

See Also

[IVertex Interface](#)
[IVertex Members](#)
[IVertex::EnumEdgesOriented Method](#)
[IVertex::GetEdgesOriented Method](#)
[IVertex::GetEdges Method](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

EnumEdgesOriented Method (IVertex)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IVertex Interface](#) : EnumEdgesOriented Method (IVertex)

Gets the enumerated edges and orients them per coedge in the direction corresponding to this vertex.

.NET Syntax

Visual Basic (Declaration)

```
Function EnumEdgesOriented() As EnumEdges
```

Visual Basic (Usage)



```
Dim instance As IVertex  
Dim value As EnumEdges
```

```
value = instance.EnumEdgesOriented()
```

C#

```
EnumEdges EnumEdgesOriented()
```

C++/CLI

```
EnumEdges^ EnumEdgesOriented();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[Enumerated edges](#)

Visual Basic for Applications (VBA) Syntax

See [Vertex::EnumEdgesOriented](#).

Remarks

The order of the edges is counter-clockwise.

See Also

[IVertex Interface](#)
[IVertex Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetAdjacentFaces Method (IVertex)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IVertex Interface](#) : GetAdjacentFaces Method (IVertex)

Gets the faces adjacent to this vertex.

.NET Syntax

Visual Basic (Declaration)

```
Function GetAdjacentFaces() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IVertex  
Dim value As System.Object  
  
value = instance.GetAdjacentFaces()
```

C#

```
System.object GetAdjacentFaces()
```

C++/CLI

```
System.Object^ GetAdjacentFaces();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of the [faces](#) adjacent to this vertex

Visual Basic for Applications (VBA) Syntax

See [Vertex::GetAdjacentFaces](#).

Example

[Get Faces Adjacent to Vertex \(VBA\)](#)

See Also

[IVertex Interface](#)
[IVertex Members](#)
[IVertex::IGetAdjacentFaces Method](#)
[IVertex::IGetAdjacentFacesCount Method](#)

Availability

SOLIDWORKS 2004 SP2, Revision Number 12.2

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetClosestPointOn Method (IVertex)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IVertex Interface](#) : GetClosestPointOn Method (IVertex)

Gets the closest point on the vertex using the X, Y, Z input point.

.NET Syntax

Visual Basic (Declaration)

```
Function GetClosestPointOn( _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _
```

```
ByVal Z As System.Double _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IVertex  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As System.Object  
  
value = instance.GetClosestPointOn(X, Y, Z)
```

C#

```
System.object GetClosestPointOn(  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI

```
System.Object^ GetClosestPointOn(  
& System.double X,  
& System.double Y,  
& System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X
X value of the input point

Y
Y value of the input point

Z
Z value of the input point

Return Value

Array of five doubles representing the X, Y, Z point on the vertex

Visual Basic for Applications (VBA) Syntax

See [Vertex::GetClosestPointOn](#).

Remarks

Because a vertex is a point, you should receive the X,Y,Z location for the actual vertex.

See Also

[IVertex Interface](#)
[IVertex Members](#)
[IVertex::GetClosestPointOn Method](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetEdges Method (IVertex)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IVertex Interface](#) : GetEdges Method (IVertex)

Gets a list of enumerated edges corresponding to this vertex.

.NET Syntax

Visual Basic (Declaration)

```
Function GetEdges() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IVertex
Dim value As System.Object

value = instance.GetEdges()
```

C#

```
System.Object GetEdges()
```

C++/CLI

```
System.Object^ GetEdges();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of [edges](#) corresponding to this vertex

Visual Basic for Applications (VBA) Syntax

See [Vertex::GetEdges](#).

Example

[Get Edges for Vertex \(VBA\)](#)

See Also

[IVertex Interface](#)
[IVertex Members](#)
[IVertex::EnumEdges Method](#)

Availability

SOLIDWORKS 2004 SP2, Revision Number 12.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetEdgesOriented Method (IVertex)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IVertex Interface](#) : GetEdgesOriented Method (IVertex)

Gets the enumerated edges and orients them per coedge in the direction corresponding to this vertex.

.NET Syntax

Visual Basic (Declaration)

```
Function GetEdgesOriented() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IVertex
Dim value As System.Object

value = instance.GetEdgesOriented()
```

C#

```
System.Object GetEdgesOriented()
```

C++/CLI

```
System.Object^ GetEdgesOriented();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of [edges](#) for this vertex

Visual Basic for Applications (VBA) Syntax

See [Vertex::GetEdgesOriented](#).

Remarks

The order of the edges is counter-clockwise.

See Also

[IVertex Interface](#)
[IVertex Members](#)
[IVertex::EnumEdgesOriented Method](#)

Availability

SOLIDWORKS 2004 SP2, Revision Number 12.2

SOLIDWORKS API Help

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GetPoint Method (IVertex)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IVertex Interface](#) : GetPoint Method (IVertex)

Gets the point corresponding to this vertex.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPoint() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IVertex
Dim value As System.Object

value = instance.GetPoint()
```

C#

```
System.object GetPoint()
```

C++/CLI

```
System.Object^ GetPoint();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of three doubles representing the x, y, and z coordinates of the point

Visual Basic for Applications (VBA) Syntax

See [Vertex::GetPoint](#).

Example

[Get Table-driven Pattern Properties \(VBA\)](#)
[Get Vertex \(VBA\)](#)

[Get Sketch Regions \(VBA\)](#)
[Get Sketch Regions \(VB.NET\)](#)
[Get Sketch Regions \(C#\)](#)

See Also

[IVertex Interface](#)
[IVertex Members](#)
[IVertex::IGetPoint Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetTrackingIDs Method (IVertex)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IVertex Interface](#) : GetTrackingIDs Method (IVertex)

Gets the [tracking IDs assigned to this vertex](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetTrackingIDs( _  
    ByVal TrackingCookie As System.Integer, _  
    ByRef TrackingIDs As System.Object _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IVertex  
Dim TrackingCookie As System.Integer  
Dim TrackingIDs As System.Object  
Dim value As System.Integer  
  
value = instance.GetTrackingIDs(TrackingCookie, TrackingIDs)
```

C#

```
System.int GetTrackingIDs(  
    System.int TrackingCookie,  
    out System.object TrackingIDs  
)
```

C++/CLI

```
System.int GetTrackingIDs(  
& System.int TrackingCookie,  
& [Out] System.Object^ TrackingIDs  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TrackingCookie
Cookie obtained from [ISldWorks::RegisterTrackingDefinition](#)
TrackingIDs
Array of tracking IDs on this vertex

Return Value

Status as defined by [swTrackingIDError e](#)

Visual Basic for Applications (VBA) Syntax

See [Vertex::GetTrackingIDs](#).

Example

[Get, Set, and Remove Tracking IDs on Vertex \(VBA\)](#)

Remarks

See [Tracking IDs](#) for more information.

See Also

[IVertex Interface](#)
[IVertex Members](#)
[IVertex::GetTrackingIDsCount Method](#)
[IVertex::IGetTrackingIDs Method](#)
[IVertex::RemoveTrackingID Method](#)
[IVertex::SetTrackingID Method](#)
[IModelDocExtension::FindTrackedObjects Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetTrackingIDsCount Method (IVertex)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IVertex Interface](#) : GetTrackingIDsCount Method (IVertex)

Gets the number of tracking IDs assigned to this vertex.

.NET Syntax

Visual Basic (Declaration)

```
Function GetTrackingIDsCount( _  
    ByVal TrackingCookie As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IVertex  
Dim TrackingCookie As System.Integer  
Dim value As System.Integer  
  
value = instance.GetTrackingIDsCount(TrackingCookie)
```

C#

```
System.int GetTrackingIDsCount(  
    System.int TrackingCookie  
)
```

C++/CLI

```
System.int GetTrackingIDsCount(  
& System.int TrackingCookie  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TrackingCookie
Cookie obtained from [ISldWorks::RegisterTrackingDefinition](#)

Return Value

Number of tracking IDs on this vertex

Visual Basic for Applications (VBA) Syntax

See [Vertex::GetTrackingIDsCount](#).

Example

[Get, Set, and Remove Tracking IDs on Vertex \(VBA\)](#)

Remarks

Call this method before calling [IVertex::IGetTrackingIDs](#) to determine the size of the array for that method.
See [Tracking IDs](#) for more information.

See Also

[IVertex Interface](#)
[IVertex Members](#)
[IVertex::GetTrackingIDs Method](#)
[IVertex::RemoveTrackingID Method](#)
[IVertex::SetTrackingID Method](#)
[IModelDocExtension::FindTrackedObjects Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetAdjacentFaces Method (IVertex)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IVertex Interface](#) : IGetAdjacentFaces Method (IVertex)

Gets the faces adjacent to this vertex.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetAdjacentFaces( _  
    ByVal NFaceCount As System.Integer _  
) As Face2
```

Visual Basic (Usage)



```
Dim instance As IVertex  
Dim NFaceCount As System.Integer  
Dim value As Face2  
  
value = instance.IGetAdjacentFaces(NFaceCount)
```

C#

```
Face2 IGetAdjacentFaces(  
    System.int NFaceCount  
)
```

C++/CLI

```
Face2^ IGetAdjacentFaces(  
& System.int NFaceCount  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NFaceCount
Number of adjacent faces

Return Value

- in-process, unmanaged C++: Pointer to an array of [faces](#) adjacent to this vertex
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Before calling this method, call [IVertex::IGetAdjacentFacesCount](#) to get the number of faces adjacent to this vertex.

See Also

[IVertex Interface](#)
[IVertex Members](#)
[IVertex::GetAdjacentFaces Method](#)

Availability

SOLIDWORKS 2004 SP2, Revision Number 12.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetAdjacentFacesCount Method (IVertex)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IVertex Interface](#) : IGetAdjacentFacesCount Method (IVertex)

Gets the number of faces adjacent to this vertex.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetAdjacentFacesCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IVertex  
Dim value As System.Integer  
  
value = instance.IGetAdjacentFacesCount()
```

C#

```
System.int IGetAdjacentFacesCount()
```

C++/CLI

```
System.int IGetAdjacentFacesCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of faces adjacent to this vertex

Visual Basic for Applications (VBA) Syntax

See [Vertex::IGetAdjacentFacesCount](#).

Remarks

Call this method before calling [IVertex::IGetAdjacentFaces](#).

See Also

[IVertex Interface](#)
[IVertex Members](#)
[IVertex::GetAdjacentFaces Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetClosestPointOn Method (IVertex)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IVertex Interface](#) : IGetClosestPointOn Method (IVertex)

Gets the closest point on the vertex using the X, Y, Z input point.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetClosestPointOn( _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
    ) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IVertex  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As System.Double  
  
value = instance.IGetClosestPointOn(X, Y, Z)
```

C#

```
System.double IGetClosestPointOn(  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI

```
System.double IGetClosestPointOn(  
& System.double X,  
& System.double Y,  
& System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X
X value of the input point

Y
Y value of the input point

Z
Z value of the input point

Return Value

- in-process, unmanaged C++: Pointer to an array of five doubles representing the X, Y, Z point on the vertex
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

See Also

[IVertex Interface](#)
[IVertex Members](#)
[IVertex::GetClosestPointOn Method](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetPoint Method (IVertex)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IVertex Interface](#) : IGetPoint Method (IVertex)

Gets the point corresponding to this vertex.

.NET Syntax

```
Function IGetPoint() As System.Double
```



```
Dim instance As IVertex
Dim value As System.Double

value = instance.IGetPoint()
```

```
System.double IGetPoint()
```

```
System.double IGetPoint();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of three doubles representing the x, y, and z coordinates of the point
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

See Also

[IVertex Interface](#)
[IVertex Members](#)
[IVertex::GetPoint Method](#)

IGetTrackingIDs Method (IVertex)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IVertex Interface](#) : IGetTrackingIDs Method (IVertex)

Gets the [tracking IDs assigned to this vertex](#).

.NET Syntax

```
Function IGetTrackingIDs( _
    ByVal TrackingCookie As System.Integer, _
    ByVal Count As System.Integer, _
    ByRef TrackingIDs As System.Integer _
) As System.Integer
```



```
Dim instance As IVertex
Dim TrackingCookie As System.Integer
Dim Count As System.Integer
Dim TrackingIDs As System.Integer
Dim value As System.Integer

value = instance.IGetTrackingIDs(TrackingCookie, Count, TrackingIDs)
```

```
System.int IGetTrackingIDs(
    System.int TrackingCookie,
    System.int Count,
    out System.int TrackingIDs
)
```

```
System.int IGetTrackingIDs(
& System.int TrackingCookie,
& System.int Count,
& [Out] System.int TrackingIDs
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TrackingCookie

Cookie obtained from [ISldWorks::RegisterTrackingDefinition](#)

Count

Number of tracking IDs on this vertex

TrackingIDs

• in-process, unmanaged C++: Pointer to an array of tracking IDs on this vertex

- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Return Value

Status as defined by [swTrackingIDError_e](#)

Remarks

Before calling this method, call [IVertex::GetTrackingIDsCount](#) to get Count.

See [Tracking IDs](#) for more information.

See Also

[IVertex Interface](#)

[IVertex Members](#)

[IVertex::GetTrackingIDs Method](#)

[IVertex::RemoveTrackingID Method](#)

[IVertex::SetTrackingID Method](#)

[IModelDocExtension::FindTrackedObjects Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

IsTolerant Method (IVertex)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IVertex Interface](#) : IsTolerant Method (IVertex)

Gets whether a vertex is tolerant and its tolerance value.

.NET Syntax

Visual Basic (Declaration)

```
Function IsTolerant( _
    ByRef Tolerance As System.Double _
) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IVertex
Dim Tolerance As System.Double
Dim value As System.Boolean
```

```
value = instance.IsTolerant(Tolerance)
```

C#

```
System.bool IsTolerant(
    out System.double Tolerance
)
```

C++/CLI



```
System.bool IsTolerant(  
& [Out] System.double Tolerance  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Tolerance
Vertex tolerance or gap in meters

Return Value

True if tolerant, false if not tolerant

Visual Basic for Applications (VBA) Syntax

See [Vertex::IsTolerant](#)

Example

[Get Maximum Edge and Vertex Gaps \(VBA\)](#)
[Get Maximum Edge and Vertex Gaps \(VB.NET\)](#)
[Get Maximum Edge and Vertex Gaps \(C#\)](#)

Remarks

If the tolerance (or gap) between common vertices in a part:

- is greater than 1/2 of the session precision (i.e., > 5.0E-9 mm), then each vertex is tolerant, and this method returns true.
- is less than or equal to 1/2 of the session precision (i.e., <= 5.0E-9 mm), then each vertex is exact and not tolerant, and this method returns false.

Traverse part body vertices and use this method to find the maximum vertex gap (tolerance) in a part, which can also be found using the **Tools** : **Check** dialog in the SOLIDWORKS Design user interface.

See Also

[IVertex Interface](#)
[IVertex Members](#)
[IEdge::IsTolerant](#)

Availability

SOLIDWORKS 2009 SP2, Revision Number 17.2.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

RemoveTrackingID Method (IVertex)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IVertex Interface](#) : RemoveTrackingID Method (IVertex)

Removes a [tracking ID assigned to this vertex](#).

.NET Syntax

Visual Basic (Declaration)

```
Function RemoveTrackingID(  
    ByVal TrackingCookie As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IVertex  
Dim TrackingCookie As System.Integer  
Dim value As System.Integer  
  
value = instance.RemoveTrackingID(TrackingCookie)
```

C#

```
System.int RemoveTrackingID(  
    System.int TrackingCookie  
)
```

C++/CLI

```
System.int RemoveTrackingID(  
& System.int TrackingCookie  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TrackingCookie
Cookie obtained from [ISldWorks::RegisterTrackingDefinition](#)

Return Value

Status as defined by [swTrackingIDError_e](#)

Visual Basic for Applications (VBA) Syntax

See [Vertex::RemoveTrackingID](#).

Example

[Get, Set, and Remove Tracking IDs on Vertex \(VBA\)](#)

Remarks

See [Tracking IDs](#) for more information.

See Also

[IVertex Interface](#)
[IVertex Members](#)
[IVertex::GetTrackingIDs Method](#)
[IVertex::SetTrackingID Method](#)
[IVertex::GetTrackingIDsCount Method](#)
[IVertex::IGetTrackingIDs Method](#)
[IModelDocExtension::FindTrackedObjects Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetTrackingID Method (IVertex)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IVertex Interface](#) : SetTrackingID Method (IVertex)

Assigns a tracking ID to this vertex.

.NET Syntax

Visual Basic (Declaration)

```
Function SetTrackingID( _  
    ByVal TrackingCookie As System.Integer, _  
    ByVal TrackingID As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)

```
Dim instance As IVertex  
Dim TrackingCookie As System.Integer  
Dim TrackingID As System.Integer
```



```
Dim value As System.Integer
```

```
value = instance.SetTrackingID(TrackingCookie, TrackingID)
```

C#

```
System.int SetTrackingID(  
    System.int TrackingCookie,  
    System.int TrackingID  
)
```

C++/CLI

```
System.int SetTrackingID(  
& System.int TrackingCookie,  
& System.int TrackingID  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TrackingCookie
Cookie obtained from [ISldWorks::RegisterTrackingDefinition](#)

TrackingID
Application-defined value for the tracking ID

Value	Action
>0	Vertex is tracked
0	Tracking of vertex is stopped

Return Value

Status as defined by [swTrackingIDError_e](#)

Visual Basic for Applications (VBA) Syntax

See [Vertex::SetTrackingID](#).

Example

[Get, Set, and Remove Tracking IDs on Vertex \(VBA\)](#)

Remarks

You can assign tracking IDs to [bodies](#), [edges](#), [faces](#), [loops](#), and vertices in parts and assemblies only; you cannot assign tracking IDs to these entities in drawings. See [Tracking IDs](#) for more information.

See Also

[IVertex Interface](#)
[IVertex Members](#)
[IVertex::GetTrackingIDs Method](#)
[IVertex::RemoveTrackingID Method](#)
[IVertex::IGetTrackingIDs Method](#)
[IVertex::GetTrackingIDsCount Method](#)
[IModelDocExtension::FindTrackedObjects Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0